AMA572 Lecture 5

Intensity Based Reduced Form Models and Introduction to Credit Derivatives

Unlike previous structural models, such as Merton, Black-Cox, and Leland-Toft models, which assume that there is a fundamental asset value which triggers the default, reduced form models do not model default explicitly. More precisely, reduced form models assume that there is some exogenous processes, such as intensity processes, drive the defaults, and take the bond prices as given. After we calibrate the defaultable bond price to the market, then we can use the bonds prices to compute prices for other derivatives. This is similar to the Black-Scholes model, in which we use stock prices to compute option prices.

Popular reduced form models include Duffie-Singleton model, which is based on intensities, and Jarrow-Turnbull model, which is based on credit rating migration. Here we shall focus on the first model.

1 Duffie-Singleton Reduced Form Model

Reduced-form models take the probability of default as given by the market, unlike the structural models (such as Merton, Black-Cox, Leland-Toft models). In this lecture we shall focus on one particular reduced-form model, Duffie-Singleton (1999) Model.

1.1 Hazard Rate and Intensity

The hazard rate $h(t,T)$ measures the conditional probability of defaulting between T and $T+dt$, given all the information available at time t. More precisely, let the default time be τ

$$S(t,T) = P(\tau > T|\mathcal{F}_t),$$

$$f(t,T) = \frac{\partial S(t,T)}{\partial T},$$

$$h(t,T) = \frac{f(t,T)}{S(t,T)} = \frac{1}{S(t,T)}\frac{\partial S(t,T)}{\partial T}.$$

Then, it can be shown that

$$S(t,T) = \exp\left\{-\int_t^T h(t,s)ds\right\}.$$

The intensity $\lambda(t)$ measures the instantaneous defaults rate. More precisely, there is a unique stochastic process $\lambda(t)$ such that

$$N(t) - \int_0^t \lambda(s)ds$$

is a martingale, and under suitable conditions

$$\lambda(t) = h(t,t).$$

Here $N(t)$ is a counting process

$$N(t) = I\{\tau \leq t\}.$$

1.2 The Model

Consider a European contingent claim whose payoff is given by $g(T)$. This is a quite general framework. For example, for the zero coupon bond $g(T) = 1$. Suppose the recovery value is a fraction, δ, of the market value of the contingent claim right before the default. The key result of Duffie-Singleton model is that the value of the contingent claim at time t is given by

$$g(t) = E^*\left[\exp\left\{-\int_t^T [r(s) + (1-\delta)\lambda(s)]\, ds\right\} \cdot g(T)|\mathcal{F}_t\right]. \qquad (1)$$

Remarks. (1) In the above formula the risk-free rate $r(s)$ may be stochastic, and the default intensity may also be a stochastic process. (2) For zero-coupon bond we have at time 0

$$B(0,T) = E^*\left[\exp\left\{-\int_0^T [r(s) + (1-\delta)\lambda(s)]\, ds\right\}\right].$$

(3) After we calibrate the default intensity $\lambda(t)$, then we can use $\lambda(t)$ to price other contingent claims. In other words, we view bond prices as given, and then use the bond prices to compute prices for other derivatives. This is similar to the Black-Scholes model, in which we use stock prices to compute option prices.

1.3 A Heuristic Derivation

We shall only give a heuristic explanation of (1). Consider a discrete time framework. We can think $h(t)dt$ as approximately the default probability in $(t, t+dt)$ and $r(t)$ as the approximate risk-free yield from t to $t+dt$. Then the standard option pricing theory implies that

$$g(t) = h(t)dt \cdot e^{-r(t)dt} E^*[\delta g(t+dt)|\mathcal{F}_t] + (1 - h(t)dt)e^{-r(t)dt} E^*[g(t+dt)|\mathcal{F}_t],$$

because the default payment is $\delta g(t+dt)$ and the non-default payment is $g(t+dt)$.

Introduce $R(t)$ such that

$$\exp\{-R(t)\cdot dt\} = h(t)dt\cdot e^{-r(t)dt}\delta + (1-h(t)dt)e^{-r(t)dt}. \tag{2}$$

Then

$$g(t) = \exp\{-R(t)\cdot dt\}E^*[g(t+dt)|\mathcal{F}_t].$$

Iterating the above expression yields

$$g(t+dt) = \exp\{-R(t+dt)\cdot dt\}E^*[g(t+2dt)|\mathcal{F}_{t+dt}],$$

and therefore

$$\begin{aligned} g(t) &= \exp\{-R(t)\cdot dt\}E^*[g(t+dt)|\mathcal{F}_t] \\ &= \exp\{-R(t)\cdot dt\}\cdot E^*\left\{\exp\{-R(t+dt)\cdot dt\}\cdot E^*[g(t+2dt)|\mathcal{F}_{t+dt}]|\mathcal{F}_t\right\} \\ &= E^*\left\{E^*[\exp\{-\left(R(t)+R(t+dt)\right)\cdot dt\}\cdot g(t+2dt)|\mathcal{F}_{t+dt}]|\mathcal{F}_t\right\} \\ &= E^*[\exp\{-\left(R(t)+R(t+dt)\right)\cdot dt\}\cdot g(t+2dt)|\mathcal{F}_t]. \end{aligned}$$

Thus, continuing this procedure, we have

$$g(t) = E^*[\exp\{-\left(R(t)+R(t+dt)+\cdots+R(T)\right)\cdot dt\}\cdot g(T)|\mathcal{F}_t]. \tag{3}$$

Now we need to have an approximation for $R(t)$. Using the approximation $e^x \approx 1+x$, we have

$$\exp\{-R(t)\cdot dt\} \approx 1 - R(t)\cdot dt,$$

$$\lambda(t)dt\cdot e^{-r(t)dt}\delta + (1-\lambda(t)dt)e^{-r(t)dt} \approx \lambda(t)dt\cdot(1-r(t)dt)\delta + (1-\lambda(t)dt)(1-r(t)dt),$$

and, therefore, we have from (2) that

$$1 - R(t)\cdot dt \approx \lambda(t)dt\cdot(1-r(t)dt)\delta + (1-\lambda(t)dt)(1-r(t)dt).$$

In summary, we have

$$1 - R(t)\cdot dt \approx \lambda(t)\delta dt - \lambda(t)r(t)\delta\left(dt\right)^2 + 1 - \lambda(t)dt - r(t)dt + \lambda(t)r(t)\left(dt\right)^2.$$

Ignoring the term $(dt)^2$ we have

$$-R(t)\cdot dt \approx \lambda(t)\delta dt - \lambda(t)dt - r(t)dt,$$

or

$$R(t) \approx \lambda(t)(1-\delta) + r(t).$$

This in conjunction with (3) gives, as $dt \to 0$,

$$\begin{aligned} g(t) &= E^*\left[\exp\{-\int_t^T R(s)ds\}\cdot g(T)|\mathcal{F}_t\right] \\ &= E^*\left[\exp\{-\int_t^T \left[\lambda(s)(1-\delta)+r(s)\right]ds\}\cdot g(T)|\mathcal{F}_t\right], \end{aligned}$$

which shows the result.

2 Credit Default Swaps

2.1 Background

A credit default swap (CDS) is a traded insurance contract that provides payments against the risk of default. The buyer of the CDS makes periodic payment to the seller until either the contract reaches maturity or a credit event occurs. In the credit event, the seller of the CDS will make certain payment. For example, a buyer may pay 90 basis point on a notation amount $\$10M$ once per year (i.e. \$90K per year) for 5 years, in exchange for a protection payment of difference between the face value of the debt and the market value of the debt after the credit event. For example, if the face value of the debt is \$100, but the market value is \$40 after the credit event, and the seller of the CDS will pay \$6m to the buyer.

Back in 2008, all CDS contracts are not traded in exchanges. Now CDS contracts are traded in several exchanges. Even individual investors can buy CDS contracts via, e.g., International Exchange (ICE) and Chicago Mercantile Exchange (CME).

Two popular indices for CDS are: 5 year and 10 year CDX NA IG index which tracks 125 investment grade North American companies; 5 year and 10 year iTraxx Europe index which tracks 125 investment grade European companies.

In September 2009 US government bailed American International Group (AIG) out, mainly because AIG had been excessively selling CDS protection without hedging against the possibility that the reference entities, especially on Lehman, might decline in value, which exposed the insurance giant to potential losses over \$100 billion.

In April 2012, the market in credit default swaps was possibly being affected by the activities of Bruno Iksil, a trader for J.P. Morgan Chase & Co., referred to as "the London whale" in reference to the huge positions he was taking. Heavy opposing bets to his positions were made by traders, including another branch of J.P. Morgan, who purchased the derivatives offered by J.P. Morgan in such high volume. \$2 billion losses were reported by the firm in May, 2012 in relationship to these trades.

2.2 Valuation

We will illustrate the valuation of 5 year CDS via an example. In this example, we shall find out the break-even CDS spread s.

Assume:

1. The default intensity during any year is 0.02.

2. The risk-free rate is 5%

3. The recovery rate is 40%.

4. Default can only happen at year 0.5, 1.5, 2.5, 3.5, 4.5.

Since the payment from the buyer is typically in arrear (e.g. at year 1, 2, ..., 5), the calculation relies on computation of 3 payments.

1. Regular payments by the buyer.

2. Accrual payment by the buyer after the credit event.

3. The payment from the seller to the buyer after the credit event.

The following table details item 1, with s being the CDS spread.

Time (Years)	Prob. of Survival	Expected Payments	Discount Factor	PV of Exp Payment
1	0.9800	$0.9800s$	0.9512	$0.9322s$
2	0.9604	$0.9604s$	0.9048	$0.8690s$
3	0.9412	$0.9412s$	0.8607	$0.8101s$
4	0.9224	$0.9224s$	0.8187	$0.7552s$
5	0.9039	$0.9039s$	0.7788	$0.7040s$
Total				$4.0704s$

For example, in Year 3, the survival probability is

$$(1 - 0.02)^3 = 0.9412;$$

the discounted factor is

$$e^{-0.05*3} = 0.860\,7;$$

and the PV of expected payment is

$$0.9412s * 0.860\,7 = 0.810\,1s.$$

The following table details item 2, with s being the CDS spread.

Time (Years)	Prob. of Default	Expected Payments	Discount Factor	PV of Exp Payment
0.5	0.0200	$0.0100s$	0.9753	$0.0097s$
1.5	0.0196	$0.0098s$	0.9277	$0.0091s$
2.5	0.0192	$0.0096s$	0.8825	$0.0085s$
3.5	0.0188	$0.0094s$	0.8395	$0.0079s$
4.5	0.0184	$0.0092s$	0.7985	$0.0074s$
Total				$0.0426s$

For example, the default probability during the third year (which is assumed to happen exactly at time 2.5 year) is

$$0.02 * (1 - 0.02)^2 = 0.0192;$$

and the corresponding arrear payment is

$$(2.5 - 2) * 0.0192s = 0.009\,6s.$$

The discount factor is

$$e^{-0.05*2.5} = 0.882\,50.$$

The following table details item 3, with s being the CDS spread.

Time (Years)	Prob. of Default	Expected Payments	Discount Factor	PV of Exp Payment
0.5	0.0200	0.0120	0.9753	0.0117
1.5	0.0196	0.0118	0.9277	0.0109
2.5	0.0192	0.0115	0.8825	0.0102
3.5	0.0188	0.0113	0.8395	0.0095
4.5	0.0184	0.0111	0.7985	0.0088
Total				0.0511

For example, in the above table, if the default happens at time 2.5, then the expected payment is

$$0.0192 * (1 - 0.4) = 0.011\,5.$$

The break-even CDS spread is solved by matching the payments between the buyer and seller, i.e.,

$$4.0704s + 0.0426s = 0.0511,$$

which yields

$$s = 1.24\%,$$

or 124 basis points per year, i.e. \$124,000 per year on the \$10M contact.

Note (i) Market CDS spread may be different from the break-even CDS spread, which may due to the different default intensities being used. In this case, traders may want to enter a trade, either on the long or short side of CDS. (2) Even if the traders agree on the initial CDS spread at time 0, at time t, the CDS spread may be considerably different from the spread at time 0, resulting in a profit/loss for one party.

HWK:

1. Suppose that the risk-free rate is at 7% per year with continuous compounding and that defaults can occur half way through each year in a new 5-year credit default swap. Suppose that the recovery rate is 30% and the default intensity each year is 3%. What is the break-even spread for the credit default swap? Assume payments are made annually.